
Scale-Down Models and Sensors for Process Development

The design and development of perfusion cultures require extensive experimental campaigns in order to identify the most convenient conditions in terms of a very large number of operating parameters. This effort can be alleviated through the use of mathematical models, which are discussed in Chapter 6. Nevertheless, there is a need for high-throughput technologies that allow for the simultaneous operation and monitoring of several cultures at different process conditions. This requires suitable scale-down models, which allow the reliable prediction of the behaviour at larger scale. Within this chapter, we describe systems that allow the operation at the μL and mL scale and that enable early process development, such as clone and media screening, as well as the design of suitable process operating conditions at a larger scale. Managing such a large number of experiments is not possible without the proper level of automation, including suitable sensors for process monitoring.

It is to be said that these technologies, in the frame of perfusion process development, are still very young, certainly far from the level of maturity and therefore reliability of the corresponding scale-down models used for batch or fed-batch bioreactors. Accordingly, sometimes in this chapter, we discuss what would be desirable to have more than what is actually available.

3.1 PERFUSION CELL CULTURE AT DIFFERENT SCALES

Scale-down models are experimental devices used at the laboratory scale to reproduce the behaviour of larger units. In cell culture, the manufacturing scale ranges from tens to thousands of litres, whereas the scale-down models range from nanolitres to litres. The use of such devices is quite common in the biopharmaceutical industry to gather information needed in various moments of the development and operation of an industrial cell culture. This requires the screening of a large number of process conditions, such as for clones or media selection, investigation of various operating conditions, such as in process optimisation or process characterisation – e.g., during validation procedures.

The typical approach in the design of chemical and biotechnology processes is to first develop them in small units and then scale them up to a larger scale, which in biopharmaceutical industry means clinical and eventually commercial applications. This is done using either mathematical models or laboratory or pilot scale units or a combination of both, depending upon their convenience and accuracy. With respect to perfusion processes, the first step of this process is the identification of suitable process conditions enabling a stable culture, with an appropriate host cell line and

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media able to express the desired protein. For this, we use scale-down models (Long et al. 2014), whose results can then be reproduced at a larger scale. These devices, also often referred to as high-throughput units, derive their name from their ability to reproduce the behaviour of larger units.

In general, there are several requirements that need to be fulfilled. On the one hand, the model must be accurate enough to reproduce the larger scale fermentation process. On the other hand, it should offer a sufficiently high throughput and require a reasonably low workload. Mammalian cell cultures typically take days or weeks to be completed, and therefore, the model must be reliable and robust to minimise failures over long-term operation. In addition, cultures must be kept in a sterile environment to avoid any contamination by foreign and undesired microorganisms.

Like in any cell culture device, in scale-down models, the environment needs to be similar to physiological conditions. For example, in bench-scale bioreactors, there is typically control of temperature, pH, dissolved oxygen and CO₂ concentration. Homogeneity of the culture environment is controlled with agitation, while other relevant operating parameters such as osmolality and nutrient concentration are controlled via media formulation, feed addition or, in the case of continuous systems, perfusion rate.

In the following, we first describe the smallest scale-down models – for example, in the nanolitre scale – with very high throughput. Next, we move to larger models but with smaller throughput and proceed to the most common scale-down model – that is, the bench-scale bioreactor – say, in the one-litre scale. Limitations and opportunities to use them as modelling or screening tools for perfusion processes are discussed with specific attention to the problem of reproducing continuous media exchange and cell retention.

3.2 NANOFUIDIC CHIPS

In Chapter 1, we introduced the challenges encountered in clone screening, when establishing a new manufacturing cell line. Berkley Lights has developed a fully integrated platform (Beacon) that uses Opto Electronic Positioning technology to perform and manipulate cells on nanofluidic culturing chips (Breinlinger et al. 2018). Common cell culture tasks can be programmed through a dedicated software, which also enables parallel maintenance and analysis of thousands of cell lines. This technology allows multiplexed deposition of single cells into an array of individual chambers with nanolitre volumes. An entire chip can be imaged in 8 minutes – including artificial intelligence-based cell counting, monitoring of clonality and growth of all clones in combination with scoring of antibody secretion by fluorescent assays. With this, the relative protein productivity can be estimated, and only clones with acceptable productivities can be selected (Goudar et al. 2017). This technology has not yet been implemented in industry for screening clones in perfusion mode, but, in principle, continuous media exchange could be applied using the constant medium flow along the channel. The interface between this laminar flow and the well would enable some exchange to replenish nutrients and evacuate toxic waste metabolites. It has been shown that this technique not only reduces the workload for the establishment of clonal cell lines but also allows effective clone selection for all kind of production processes. In particular, this system allows the selection of cells with

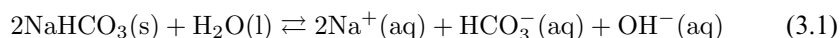
rare phenotypes or desirable traits, such as high productivity for a fed-batch process or steady growth and productivity for long-term perfusion runs (Le et al. 2018).

This technology is very specific to cell line screening, but it is discussed here because it is very innovative, and it is the only known device that could allow the cultivation of single cells with a continuous medium exchange at this scale.

3.3 INCUBATED DEVICES AS SCALE-DOWN MODELS

In this section, we consider culture devices such as Erlenmeyers, shake tubes or deepwell plates that are incubated in a controlled environment able to maintain viable culture conditions. Typically, temperature, humidity and CO₂ are controlled in the incubation environment itself and agitation is used to favour mixing and oxygen transfer to the culture.

These culture devices are closed to maintain sterility, and a gas-permeable membrane allows for oxygen and CO₂ transfer. The oxygen transport from air into the liquid suspension, where it is consumed by the cells, is controlled by the agitation in the liquid phase and the surface area of the gas-liquid interphase. Since these are both modest, oxygen supply in these devices may become a problem. Pure oxygen is sometimes used in the sparger line of benchtop bioreactors, but it is not an option for incubation devices because of safety reasons. In these devices, pH is neither monitored nor controlled, but a buffer system based on the equilibrium of sodium bicarbonate and CO₂ in air is used. The following equations describe the buffer system of sodium bicarbonate decomposing into weak acid carbonate and hydroxide while remaining in equilibrium with CO₂:



It is also worth noting that the removal of CO₂ from the culture suspension is not very efficient in these devices, since the stripping effect – characteristic of the air bubbles rising in a benchtop bioreactor – is missing. On the other hand, agitation is generally sufficient to ensure efficient heat transfer and satisfactory homogeneity of the cell suspension. The sample strategy has to be designed depending on the working volume, which may differ significantly among the different devices, as described in detail in the following section.

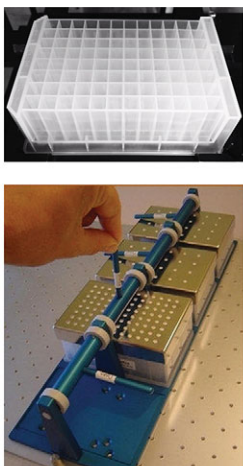
3.3.1 Shake Tubes and Deepwell Plates

Shake tubes were developed to serve as scale-down models for mammalian cell culture (De Jesus et al. 2004). They are typically 50 mL tubes designed to be used in incubated devices with agitation (Figure 3.1). The specific gas-liquid interphase surface area can be optimised by changing the culture volume in the tube and also the agitation conditions. The interphase surface area in the tube increases when the agitation is increased (Zhu et al. 2017).

Operations like sampling, seeding, feeding, etc., are done manually by an operator under a laminar flow hood to guarantee sterility. The culture volume of these models is sufficient for thorough characterisation of the cell culture in terms of cell density, metabolite concentration, osmolality, spent media analysis, product quantification

3.3 Incubated Devices As Scale-Down Models

Figure 3.1 Shake tubes of different size. The white spots on the yellow caps correspond to gas permeable membranes that keep the system sterile while maintaining in equilibrium the gas phases in the incubation chamber and inside the tube.



Deepwell plate set-up for incubation



Robotic platform for liquid handling

Figure 3.2 On the left-hand side, a 96-deepwell plate without the lid (top) and an incubator with the plate covered by the metallic lid (bottom). On the right-hand side, a robotic platform for liquid handling. This platform must be placed under a laminar flow hood to work under sterile conditions. Biomek® i7 Automated Workstation © 2017 Beckman Coulter, Inc. All rights reserved. Used with permission

and product quality evaluation. On the other hand, pH and DO are not monitored or controlled during the culture.

Besides the shake-tube technology, deepwell plates, which are very similar to traditional well plates but are deeper and therefore offer a larger culture volume, represent a promising alternative for the application in incubation devices (Figure 3.2). They exist in different formats (6–96 wells) and can be used to grow cells in suspension, similarly to shake tubes. During cultivation, the plates are covered with a lid that is permeable to gases but serves as a sterility barrier. By placing the deepwell plate into an incubator, agitation, pH and oxygen transfer are controlled just like in shake tubes. The working volume and the agitation speed must be carefully selected to guarantee sufficient oxygen transfer through the gas–liquid interface and good mixing conditions (Duetz 2007, Running & Bansal 2016).

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With such a large number of experiments running in parallel, manual handling is not feasible, and robotic platforms have to be used for liquid handling. These devices can be programmed to sample, seed and feed each plate. For example, in a 96-deepwell plate, different media can be blended independently in each well, thus creating 96 different media composition running in parallel in the same experiment (Jordan et al. 2012, Sokolov et al. 2017a). Different robotic arms have either 96 tips used, for example, to sample the entire plates at once or independent tips that can be programmed to add/remove liquids to only specific wells and with specific amounts. These devices provide, of course, a very high throughput, and are therefore best suited for media blending and screening applications. However, the reduced volume, especially in the 96-well format, may pose limitations in terms of analytics. Quality attribute characterisation often requires a substantial amount of protein that cannot always be obtained in such small volumes. If the product concentration is high enough, one can sacrifice one well by withdrawing the entire volume and then stopping the well. This is mostly done at the end of a batch or fed-batch experiment.

Many applications of deepwell plates can be found in literature. Most of them include media blending studies, such as the one of Jordan et al. (2012) on amino acid composition, where up to 192 different conditions were generated and analysed. Rouiller et al. (2013) generated 376 different conditions starting with 16 medium formulations and shuffling 43 out of the 47 media ingredients. With respect to cell line screening, it was shown that the deepwell-plate platform can be used to predict productivity and even quality attributes at larger scale, by starting with hundreds of clones and performing selection in parallel with scale-up from shake tubes, to micro-scale bioreactors and eventually to bench-scale bioreactors (Bielser et al. 2019b, Rouiller et al. 2016). Investigations on product quality were also reported (Brühlmann et al. 2017a, 2017b; Sokolov et al. 2017a). A schematic representation of the experimental workflow in these systems is shown in Figure 3.3.

The adopted analytical devices must also be adapted to this high-throughput format. For cell counting, for example, various commercial devices exist based

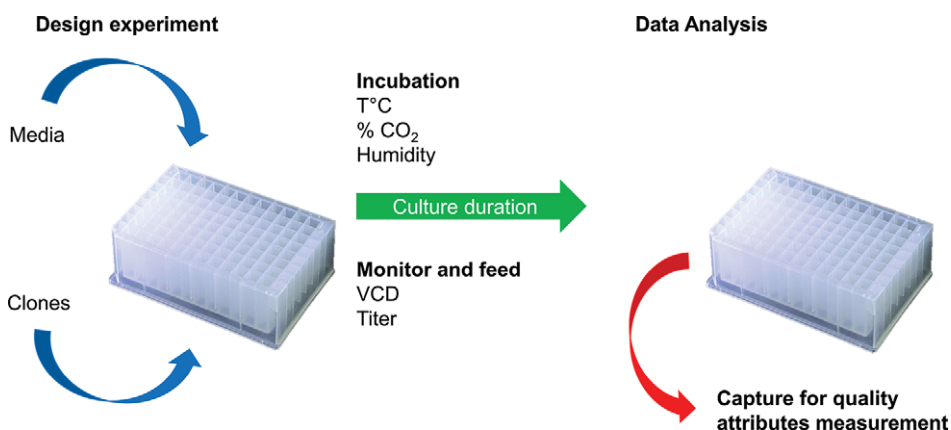


Figure 3.3 After proper experiment design, the defined amounts of media and the selected clones are added to each well of the plate. During cell culture, the plate is incubated and the robotic device is used to sample and feed the culture. High-throughput cell counting and protein titration devices are used to collect the data. At the end of the culture, the proteins can be captured for further quality attribute measurements.

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on flow cytometry technologies. For the quantification of the secreted protein, high-throughput solutions are available based on fluorescent labelling or, in particular for the case of monoclonal antibodies, on epitope binding kinetics detection.

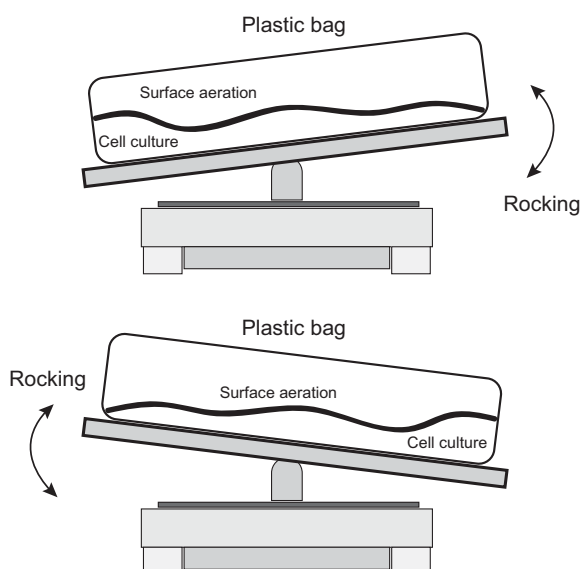
It can be noted that the format of these systems, with the handling of very small quantities, inevitably generates non-negligible variability in the final data. Therefore, we recommend using these technologies mainly for screening or performance-ranking purposes.

3.3.2 Strategies for Perfusion Cell Culture in Incubated Systems

Shake tubes and deepwell plates are well-established systems in process development for batch or fed-batch processes (Barrett et al. 2010, De Jesus et al. 2004, Gomez et al. 2017, Joao De Jesus & Wurm 2013, Rouiller et al. 2016). As mentioned in Section 1.3.3, semi-continuous operation with periodic and controlled media exchange can be used to mimic perfusion conditions in such devices. The main steps are to separate the cells from the culture suspension – for example, through sedimentation or centrifugation – and then to remove the supernatant or harvest media. This mimics the harvest stream in a real perfusion system since it contains spent medium, toxic waste metabolites and the secreted product. The removed harvest is then replaced by fresh medium that was previously conditioned, thus mimicking the perfusion stream.

Villiger-Oberbek et al. (2015) used this procedure to reach cell densities of 50×10^6 cells/mL and sustain a long-term culture for more than 58 days at about 30×10^6 cells/mL. Henry et al. (2008) predicted some parameters of the culture perfusion – such as cell growth, nutrient consumption, metabolite production and titre – using spinner flasks operated in the semi-continuous mode. Bielser et al. (2019b) used 96-DWP and ST to screen clonal cell lines, identify the best producer and predict important perfusion parameters such as CSPR and volumetric productivity. A comparison of cell-specific growth, glucose consumption, ammonia and monoclonal antibody production rates in semi-continuous shake tubes and benchtop bioreactors is discussed in detail in Section 4.4.1 (Wolf et al. 2019a). These scale-down models of perfusion cultures offer very interesting alternatives to bioreactor systems due to their much larger throughput – which makes them attractive, particularly during screening and earlier stages of process development. Of course, there are also important limitations with respect to the semi-continuous nature of the operation. For example, very high cell densities might be difficult to maintain because the required media exchange frequency might be too high. The most reasonable exchange frequency, which – keeping constant the exchange volume – is proportional to the perfusion rate, is in fact limited to once a day. Twice would require a 12 h interval, posing some organisational issues. Oxygen transfer limitations may occur in both ST and DWP systems at sufficiently large cell densities. Also, pH is not well regulated in these systems, and this can alter the product quality or even the overall culture performance (Bielser et al. 2019b, Wolf et al. 2019a). Thus summarising, although these scale-down models for perfusion are far from perfect, they can still be used not only for screening purposes but also for estimating at least a first guess of the fundamental operating parameters, such as an initial guess of the minimum CSPR, for larger scale perfusion bioreactors. Examples of applications of these systems in clone screening and process development are discussed in Chapter 4.

Figure 3.4 Principle of a wave bioreactor: the rocking movement creates waves and turbulences that favour oxygen transfer and mixing of the cell suspension. The bag is inflated with the desired gas mixture (often air and CO₂).



3.4 WAVEBAGS

Wavebag bioreactors are very traditional in cell culture and provide a category by themselves. Their name is self-explanatory: they are bags filled with culture media, lying on a flat surface that is continuously oscillating between two angles, as shown in Figure 3.4. Their operation is more similar to an incubated device than to a controlled bioreactor. The surface underneath the bags is heated and kept at a temperature value regulated through a thermometer located close to the bag. The gas mixture (air and 5 per cent CO₂ as a pH buffer) is introduced to the bag through a permeable filter. Because of the large surface-to-volume ratio, this usually provides sufficient exchange of the gaseous species.

Although wavebags have been used for perfusion cell cultures in the past, they are mostly used for seed train rather than for process-development-related activities. For example, the use of highly concentrated 5 mL cell bank vials, with densities of 90 to 100×10^6 cells/mL (Tao et al. 2011) in the seed train, instead of more traditional low-density banks, can save up to 9 days of expansion. This is extremely important from an operational point of view, since it reduces the time interval between thawing and inoculation of the production bioreactor. Wavebag bioreactors are ideal for generating high-density cell banks because the volume that they can handle is enough for a large number of vials, and this step does not require a large experimental throughput.

Some applications of wavebags for perfusion culture operation have been reported in the literature using tangential flow filtration (either TFF or ATF). Cell densities up to 130×10^6 cells/mL have been reported to stay viable for up to 20 days, with an impressive peak at 214×10^6 cells/mL (Clincke et al. 2013a). It is interesting to note that these results are made possible through the high gas–liquid mass transfer rates characteristic of these systems, which comes from the suspension surface-to-volume ratio that is very large in comparison to other cultivation devices. Foam formation is also not common in these devices. However, wave bioreactors are currently not considered for industrial production since the rocking movement that provides agitation of these bags limits their scale-up as the energy required

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must be sufficient to displace the entire culture volume, which is not feasible when considering thousands litres bags.

3.5 BIOREACTORS

A typical bioreactor system is constituted by a closed fermentation vessel and includes various devices for proper monitoring and control of the culture environment, as introduced in Chapter 2. Sterility can be preserved because the reactor is closed and all operations are performed so as to prevent contamination. To be sterile in the first place, the reactors are either autoclaved or gamma irradiated, depending on whether they are glass or single-use plastic devices. Different types of sensors monitor temperature, pH and pO_2 while control loops are in place to keep these parameters within specified intervals. In most cases, pCO_2 and other more sophisticated parameters, such as the biomass or specific nutrient levels, are also monitored and controlled. In perfusion systems, the flowrates of the media inlet, the harvest and the bleed stream are also subject to different control strategies, as already discussed in Chapter 2. In the following, we describe how the different parameters are controlled in the context of a scale-down model for the manufacturing scale.

3.5.1 Automated Microscale Bioreactors

So-called microscale bioreactors are among the smallest available scale of bioreactors on the market. They exist at two different scales, 15 and 250 mL working volume, and are operated in blocks of 24 or 48 reactors for the 15 mL scale and 6–24 for the 250 mL scale. A robotic platform can be programmed by the operator for different tasks, such as sampling and feeding, but also to monitor and control the set points of the different culture parameters. The platform can perform liquid handling using a robotic arm and sterile tips, gas lines are directly connected to the bioreactors, agitation is provided via an impeller, and the stations on which the bioreactors are located can be temperature-controlled. The mL working volume is comparably high compared to the 96-DWP or the nanofluidic chip technology, and therefore, both shake-tube and ambr[®] systems are better suited for the early stages of process development than for screening applications.

Examples of these devices are shown in Figures 3.5 and 3.6, where the vessels are in single-use format and include pH and DO monitoring probes. The advantage of micro-scale bioreactors is obviously that the experimental throughput is significantly increased compared to traditional bench-scale bioreactors. This is achieved also through a high level of automation in sampling and feeding, which does not require any human intervention.

The ambr[®] 250 is a very recent device, but it has already proven to be an efficient scale-down model. Xu et al. (2017a) report that, using the rate of gas–liquid mass transfer as a scale-down parameter, they could generate comparable cell growth and protein production profiles in the ambr[®] 250, in 5 and 1,000 L bioreactors.

In both of the equipment described in the previous paragraph, monitoring of cell density, metabolite concentrations and quality attributes is done through offline or at-line sampling. For this, the 15 mL format may not be suitable, due to volume limitations and, particularly in the case of monitoring quality attributes over time, the

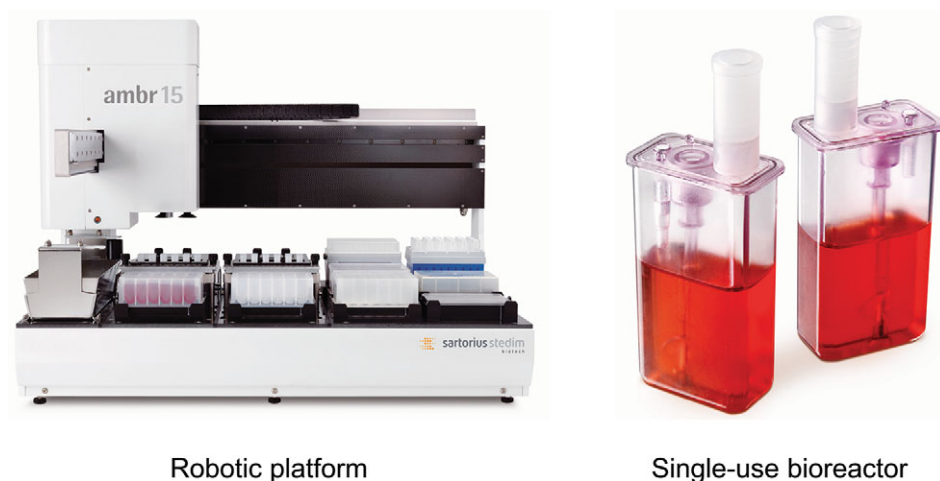


Figure 3.5 The ambr® 15 high-throughput station for 24 bioreactors (left) and schematic of the 15 mL single-use bioreactor designed for this platform, ambr® images provided courtesy of Sartorius Stedim Biotech



Figure 3.6 The ambr® 250 high-throughput station for 24 bioreactors (left) and the 250 mL single-use bioreactor used in the platform, ambr® images provided courtesy of Sartorius Stedim Biotech

sampled volumes may become significant with respect to the feed volumes. Therefore, when planning a series of experiments, like in the case of the design of an experimental program, the sample volumes required during cell culture must be verified to ensure that the maximum and minimum values are within the ranges provided by the respective equipment.

Although these systems were originally designed to mimic batch and fed-batch operations, the extension to perfusion cultures, by introducing continuous media exchange and cell retention is possible although not simple. In these platforms, the bioreactors are fixed in agitation blocks that hold twelve bioreactors each. This makes handling the bioreactors for semi-continuous operation cumbersome. However, Janoschek et al. (2018) used centrifugation, similarly to what was described earlier for shake tubes and deepwell devices, to maintain steady-state conditions at a density of 30×10^6 cells/mL for a triplicate and for about one week. An alternative, in order to replace spent media with fresh media without removing cells, would be

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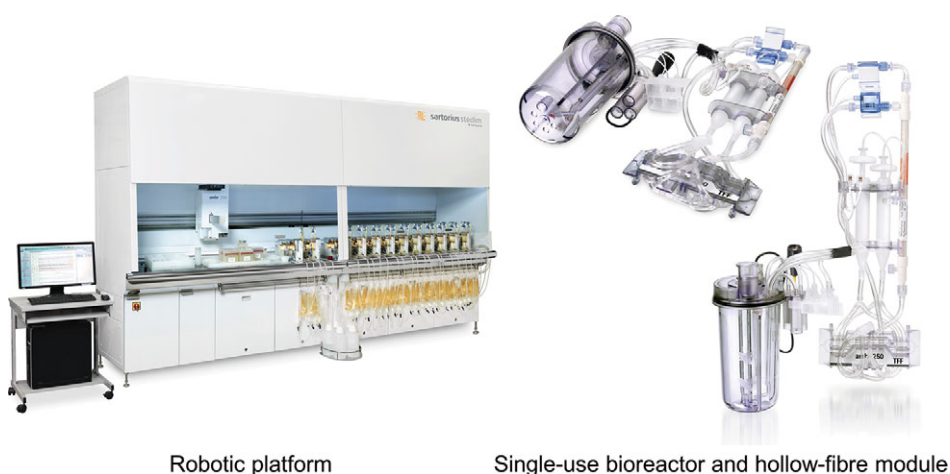


Figure 3.7 Robotic platform and single-use bioreactors for perfusion cell culture on ambr[®] 250. The bioreactors are connected to hollow fibres that can be operated either as TFF or ATF. Ambr[®] images provided courtesy of Sartorius Stedim Biotech

to let these cells settle in the bioreactor. For this, agitation and aeration should be stopped for some time until the cells settle (Davis et al. 2015, Goletz et al. 2016, Kreye et al. 2015). This can certainly lead to problems with respect to oxygen requirements, particularly in the case of higher cell densities. Nevertheless, the automated mini-bioreactor (ambr[®]) system at the 15 mL scale can be very convenient due to the high degree of automation that allows programming the sedimentation and medium exchange steps with a dedicated software. The system allows the operation at medium exchange rates of up to two reactor volumes per day and sedimentation yields cell separation efficiencies above 95 per cent.

Recognising the need for a real perfusion device, a novel type of bioreactors was developed for the ambr[®] 250. These bioreactors are equipped with hollow-fibre modules, as shown on Figure 3.7, and that can be used either in the TFF or in the ATF flow mode (Xu et al. 2017a, Zoro & Tait 2017).

3.5.2 Bench-Scale Bioreactors

Bench-scale bioreactors are glass or single-use plastic vessels in the volume range of one to ten litres. This scale is often used as a reference point for the scale-up to clinical and commercial scales, because the processes occurring at this scale are typically sufficiently similar to the corresponding ones occurring at the larger scale to allow for a reliable scale-up based on well-known engineering procedures. In a typical laboratory set-up, a central control unit receives data and sends instructions to a number of peripheral process units. Each of these controls a single vessel (Figure 3.8). Online monitoring and control typically include a temperature sensor, a pH probe, an optical dissolved oxygen probe and sometimes a dissolved carbon dioxide probe. The online data generated by these sensors are processed by the central control unit that, in turn, operates on the different process parameters.

Temperature is controlled through a heating jacket surrounding the vessel, operated through a water circuit heated externally, or an electric resistance device. In both

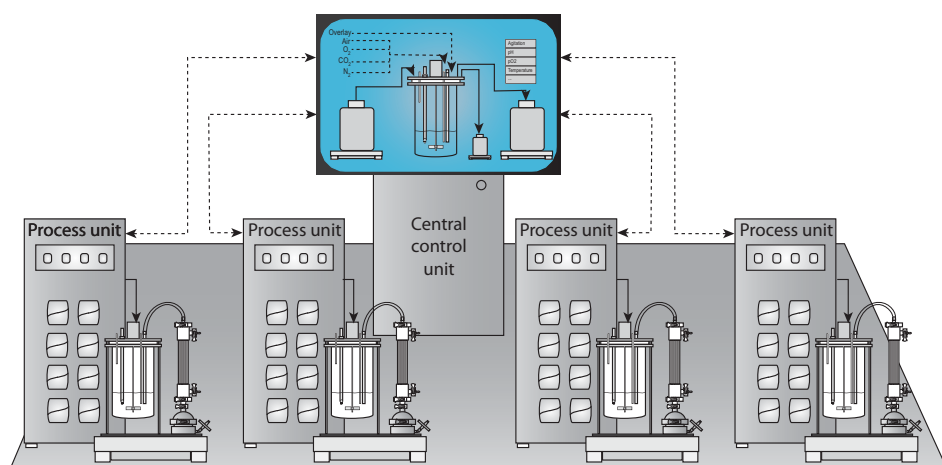
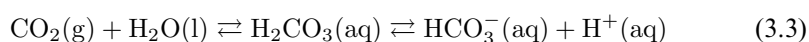


Figure 3.8 Schematic of the series of four bioreactors with a central control unit that collects online measurements and drives the different peripheral units located behind each bioreactor. These distribute the different services, such as heating/cooling water, gases, stirring and feeds or media, to the bioreactors.

cases, an efficient heat exchange requires that the liquid inside the vessel is sufficiently well agitated. This is typically the case in mammalian cell cultures since sufficient mixing is needed anyway to keep the cells in suspension and ensure homogeneity. Mixing is indeed a crucial point also considering its role on the rate of gas–liquid mass transfer. On the other hand, the shear generated by mixing can be dangerous for the cell viability. This aspect is more critical for mammalian cells than for other cell types, like bacteria or other microorganisms, due to their more delicate external membrane. However, at the scale of a few litres, a good compromise among all these factors can easily be found, but the situation becomes more critical at larger scales. For this reason, the issue of providing sufficient mixing through a proper design of the stirrer geometry and speed, without endangering the cell viability, is treated in more detail in Chapter 5.

The proper control of the bioreactor pH is essential since it impacts the cell culture viability, the cell-specific productivity and even the product quality such as its glycosylation pattern (Ivarsson et al. 2015; Monteil et al. 2016; Villiger et al. 2016a, 2016b; Yoon et al. 2005). For this, we need to have the possibility of taking two different actions: one to increase pH and the other to decrease pH. In particular, a base solution (sodium bicarbonate or sodium hydroxide) is commonly used in the first case, while the second action is performed by injecting CO₂ in the reactor, which acts on the pH through the following chemical reaction:



The pH is also affected by the metabolism of the cells, as they might produce or consume acidic/basic species over the course of the culture (Li et al. 2012, Mulukutla et al. 2012, Zagari et al. 2013).

On the other hand, since pH is such a sensitive indicator of the cell culture state, it can also be used to control the reactor operation. For example, Gagnon et al. (2011) used the pH value to control the glucose concentration in the reactor by acting on its

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feed rate. Xu & Chen (2016) reported instead the case of a cell culture that was sustained at high density and over several days without close pH control. This is probably a special feature of the specific cell line and media considered, which does not alter the conclusion that pH needs to be closely monitored and controlled in bioreactors.

Dissolved oxygen concentration in the cell culture is crucial for at least two reasons. Firstly, oxygen is consumed by the cells in the tricarboxylic acid cycle (TCA) cycle (respiration), which then cannot survive without oxygen. Secondly, high concentrations of dissolved oxygen may favour the generation of reactive oxygen species (ROS) and harm the viability of the cell culture. Therefore, the saturation of dissolved oxygen in the liquid phase of a mammalian cell culture has to be controlled at a prescribed percentage (for example 50 per cent of saturation with air). This guarantees that enough oxygen can be supplied to the cells for respiration but minimises the formation of reactive oxidating species. Dissolved oxygen can be controlled by a stream of air or pure oxygen to the culture, which keeps its level within the desired bounds.

The rate of oxygen transfer to the liquid phase depends on different factors and is quantified by the volumetric mass transfer coefficient known as $k_L a$, given by the product between the gas–liquid mass transfer coefficient, k_L , and the interphase surface area per unit reactor volume, a , multiplied by the driving force, related to the concentration difference between the liquid and the gas phase (Tribe et al. 1995). This accounts for the effect of many variables on the gas–liquid mass transfer rate. For example, going from air (≈ 21 per cent O_2) to pure oxygen increases the driving force of mass transfer, while the shape of the sparging pipe affects the interphase surface area. The latter is a consequence of the effect of the sparger type, ranging from open pipes to drilled holes or even micropores, where the size of the sparging element can vary from a few μm to a few mm, on the bubble size. This has obvious consequences on the interphase specific area which, for example, increases by 10 times if, for the same volumetric gas flow, the bubble diameter decreases from 1 mm to 0.1 mm. It is worth noting that, before inoculation or in general at very low cell densities, the gas consumption rate is very low and, therefore, the oxygen concentration in the liquid phase approaches equilibrium conditions with the gas stream. If this is too large, it can be decreased by properly diluting the gaseous stream with a second one made of pure nitrogen.

Aeration using spargers is efficient with respect to gas exchange but also has some disadvantages. When the bubbles reach the surface of the liquid, they burst, thus locally delivering some energy in the form of shear that can damage the cells. Usually, the smaller the bubble is, the larger the energy liberated and the larger the damage to the cells (Chisti 2000, Meier et al. 1999, Xu et al. 2017c). But, as discussed earlier, small bubbles are more efficient for oxygen transfer and therefore generally preferred in the case of high-cell-density bioreactors. As seen in Section 2.5, some media components such as pluronic acid can act as shear protectant for the cells and therefore strongly reduce this issue. Foam is a second side effect of large aeration rates and small bubbles. This is particularly disturbing at the small scale, where it accumulates, quickly reaches the upper region of the vessel and enters the condenser. This is, in fact, used to retain the humidity of the gaseous stream leaving the liquid suspension inside the reactor. When too much foam accumulates in the reactor, it enters the condenser and eventually reaches the filter located at its outlet. This can result in filter clogging and, as a consequence, the pressure inside the vessel increases, causing

the interruption of the operation. Different strategies can be adopted to counteract excessive foam formation. These include commercial polymers that modify the surface tension of the bubbles (Eleftherios 1991, Routledge 2012) and are typically added from the top of the bioreactor to prevent foam formation. The concentration of these compounds must be closely controlled because they can have some toxic effects. An alternative is the introduction of classical physical foam-breaking devices (Deshpande & Barigou 1999, Ishida et al. 1990, Takesono et al. 2006). At the large scale, foaming is less of an issue: sprinkler devices can be used to spray the culture surface and break the foam. The added water volume is negligible compared to the reactor volume.

3.6 SENSORS

3.6.1 In Process Control

The operations performed to periodically sample and characterise offline various parameters in the bioreactor are usually referred to as IPC. For example, viable cell density, pH, and metabolites like glucose and lactate are quantified based on a suitable sample taken on a daily basis. The obtained pH value is then used to correct possible deviations from the set-point value, while the information about cell density and viability serves to observe the progression of the culture. Metabolite concentrations like glucose, glutamine, glutamate, lactate, and ammonia are also commonly used to monitor the process. In particular, from the glucose concentration the feed addition needed to prevent its full depletion is calculated. Obviously, the methods used for offline monitoring of cell cultures are not affected by the operation mode, whether batch or continuous. We therefore do not provide here a description of the many possibilities in this direction and refer the interested reader to the specialised literature (Hu 2012, Zhou & Kantardjieff 2014).

Compared to online monitoring and control, IPC – with its daily sampling, sample preparation and characterisation – is obviously far more time consuming and less efficient, although in general can, of course, access more characteristics of the culture and the involved biomolecules. For example, considering the quantification of the metabolites, the sample needs to be first centrifuged and filtered. Other quantifications of the spent media (amino acid or vitamin concentrations for example) and critical product quality attributes (N-linked glycosylation, fragments, aggregates and charge variants) are usually performed in a single sequence only at the end of the culture run. In these cases, in fact, sample preparation takes significantly more time and cannot be performed on a daily basis. A non-exhaustive list of parameters that are measured during a typical bioreactor run and the corresponding needed sample preparation are reported Table 3.1.

More advanced online monitoring using spectroscopy techniques do exist and are continuously being developed. These can potentially replace some of the previously mentioned daily or end-of-run methods, leading to faster and more reliable process monitoring. Some of these methods are discussed in the following section.

3.6.2 Spectroscopic Sensors

Spectroscopy, based on the interaction between matter and electromagnetic radiation, provides the foundation for a number of analytical techniques using a

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Table 3.1 Parameters typically monitored during a cell culture run. Their frequency can be very different and range from online monitoring to daily sampling, to end-of-run characterisation.

Target	Parameter	Frequency	Sample preparation
<i>Bioreactor monitoring</i>	T, pO ₂ , pH, pCO ₂	Online	No sample
<i>Cell quantification and health</i>	VCD, viability, cell diameter	Daily	Possibly dilution
<i>Reactor environment</i>	pH, pCO ₂	Daily	Sample
<i>Metabolites</i>	Glucose, lactate, glutamine, glutamate, Na ⁺ , K ⁺	Daily	Filtration (0.22 µm)
<i>Spent media analysis</i>	Amino acids, vitamins, trace elements	End of run	Filtration (0.22 µm) + dilution
<i>Product quality</i>	Charge variants, N-linked glycosylation	End of run	Filtration (0.22 µm) + dilution + glycan isolation
<i>Contaminants</i>	Fragments, aggregates	End of run	Filtration (0.22 µm) + dilution

wide range of frequencies. These include ultraviolet (UV), near-infrared (NIR), mid-infrared (MIR), Raman and fluorescent spectroscopy that can be used to obtain spectral characteristics of different materials, from which information about atomic, molecular or even crystal structures of the material can be derived. Some of these techniques and the corresponding wavelengths are presented in Figure 3.9.

Different spectroscopy techniques were used to monitor online mammalian cell cultures either to measure a single quantity or, coupled with chemometric modelling techniques, multiple quantities simultaneously. Focus is given here on a few of these techniques that seem to be most promising, by discussing their potential and limitations. Most of these techniques are not yet mature enough for routine application, but due to the high potential to develop increasingly efficient process analytical technologies (PAT), particularly in the context of continuous biomanufacturing, it is not difficult to predict that they will strongly grow in the near future (Swann et al. 2017). Currently, in most cases, only temperature, dissolved O₂/CO₂ and pH are controlled online. However, using these monitoring techniques, this situation could change. For example, the feed stream could be prepared online and on-demand, based on some advanced monitoring techniques.

It is worth mentioning that the amount of information contained in such online data is very large. More complex real-time data treatments, based on multivariate data analysis (MVDA), possibly coupled with suitable mechanistic models, are needed to extract information to be directly implemented in reactor monitoring (Glassey et al. 2011). In the following, we focus on the most promising techniques with respect to continuous cultures.

Dielectric Spectroscopy. Dielectric spectroscopy, in contrast with the techniques illustrated in Figure 3.9, is based on the interaction of an external field with the

3 Scale-Down Models and Sensors for Process Development

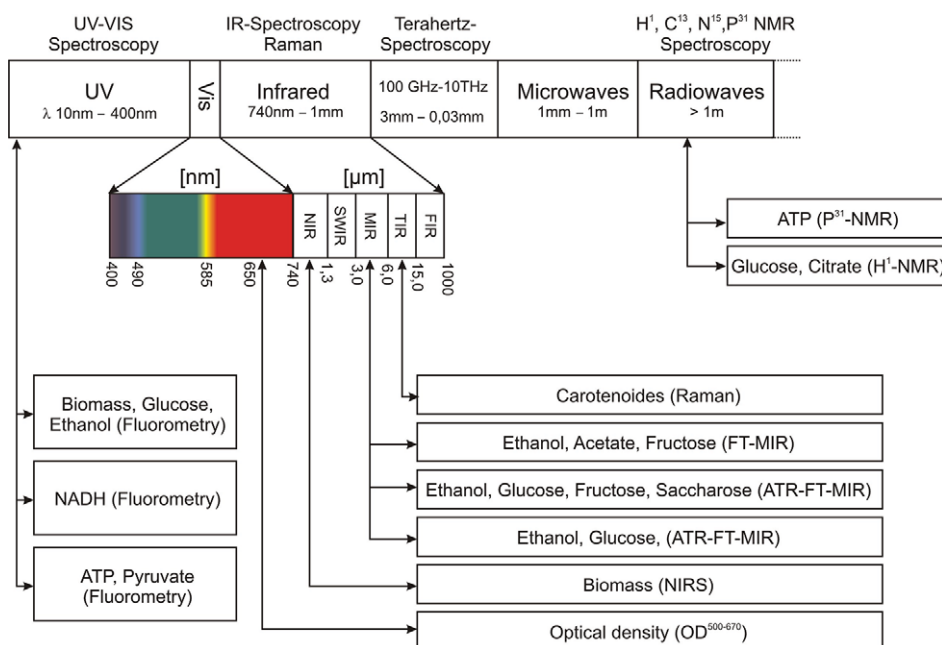


Figure 3.9 Electromagnetic wavelength spectrum and ranges of the corresponding optical and spectroscopic analytical techniques. © Reprinted by permission from *Springer Nature: Springer, Applied Microbiology and Biotechnology*, ‘In situ sensor techniques in modern bioprocess monitoring’, Beutel S. and Henkel S., 2011

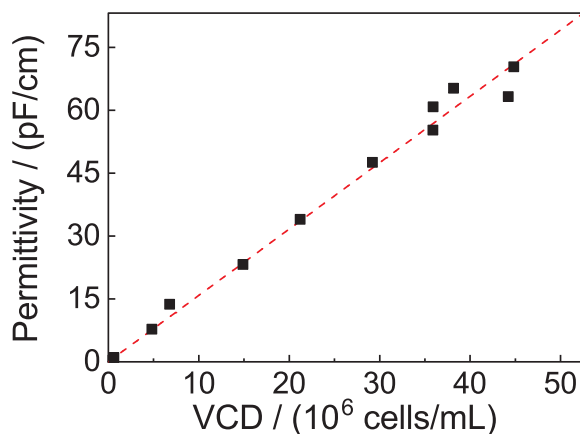
electric dipole moment of the sample; it was used to quantify the biomass even decades ago (Ducommun et al. 2001a, Joeris et al. 2002, Konstantinov et al. 1994). This technique is based on the dielectric interaction of the cells with the surrounding media as a function of frequency (Davey et al. 1993). Since the viable cells are polarised, their membrane retains some of the ions moving towards the electrodes, leading to a polarisation potential that is measured as a capacitance (pF) value. To be independent of the geometry of the probe, the capacitance is normalised into a permittivity value (pF/cm).

Only viable cells are detected using this technique, because the membrane of cells that are undergoing apoptosis or necrosis is porous and cannot be polarised. Therefore, permittivity is affected by the viable biomass, which is related to the VCD through the cell diameter d (assumed to be spherical) as follows:

$$\text{Biomass} = \frac{1}{6} \pi d^3 \times X_V, \quad (3.4)$$

but also by many other parameters in a cell culture that influence its dielectric characteristics, including some physiological changes that the cells may undergo (Opel et al. 2010, Zalai et al. 2015). Nevertheless, these sensors, proved to be very efficient to monitor viable cell densities online. This is done through the development of suitable correlations between permittivity and viable cell density, based on VCD measurements taken offline by a cell counter. In cases where the cell metabolism and, in particular, the diameter are rather constant throughout the culture, these correlations are linear, as illustrated in Figure 3.10.

Figure 3.10 Linear correlation of the permittivity signal with the offline measured viable cell density.



Dielectric spectroscopy has been used to control fed-batch bioreactors, as reported by Ansorge et al. (2010), Zalai et al. (2015) and Zhang et al. (2015a). The feed flow rates were adapted to the cell growth rate not only to reduce inter-run variability but also to improve the performance of the cultures. For perfusion cell cultures, this sensor is often used to control the bleed rate, as described by Dowd et al. (2003), Karst et al. (2016) and Warikoo et al. (2012). This real-time reaction to changes in the cell culture behaviour allows reaching a stable biomass concentrations faster.

However, it is worth pointing out that keeping constant permittivity means keeping constant the overall biomass in the reactor. This is equivalent to controlling the viable cell density only if the cell size remains constant, and, therefore, the correlation between the two, such as the linear correlation shown in Figure 3.10, stays unchanged throughout the entire culture.

The following example provides a very clear illustration of this concept (Bielser et al. 2019b). Two clones, producing the same monoclonal antibody, were cultivated in a 26-day long-term perfusion run by maintaining a constant permittivity signal in the bioreactor. In the two runs, the cells were inoculated at a similar VCD of 0.5×10^6 cells/mL (Figure 3.11A). The perfusion was started after 3 days at a targeted value of 1.0 RV/day, and the bleed was activated as soon as the permittivity set point of 70 pF/cm was reached (Figure 3.11B), which happened for both clones at day 11. For Clone 1, the VCD at this point was around 63×10^6 cells/mL, while for Clone 2, it was around 40×10^6 (Figure 3.11A). On the other hand, the cell diameter was equal to 14.7 μm for the first one and 16.5 μm for the second (Figure 3.11C). Note that both these values have been measured offline, and from these, the biomass percentage in the reactor shown in Figure 3.11B has been computed using Equation (3.4). Thus, we see that having reached the same permittivity value, the two cultures exhibit the same biomass, although different VCD (and cell diameter) values. After day 11, as expected, due to the control action, the permittivity remained constant 70 pF/cm for both clones. However, the behaviour of the VCD was significantly different. In particular, for Clone 1, the VCD decreased slowly from 63 to 42×10^6 cells/mL between day 11 and 17 and then remained stable for the rest of the culture. For Clone 2, instead, the VCD remained constant and stable at around 40×10^6 cells/mL until the end of the culture. Similarly to the VCD, the behaviour of the cell size was also different for the two clones. For Clone 1, the cell diameter increased from 14.7 μm

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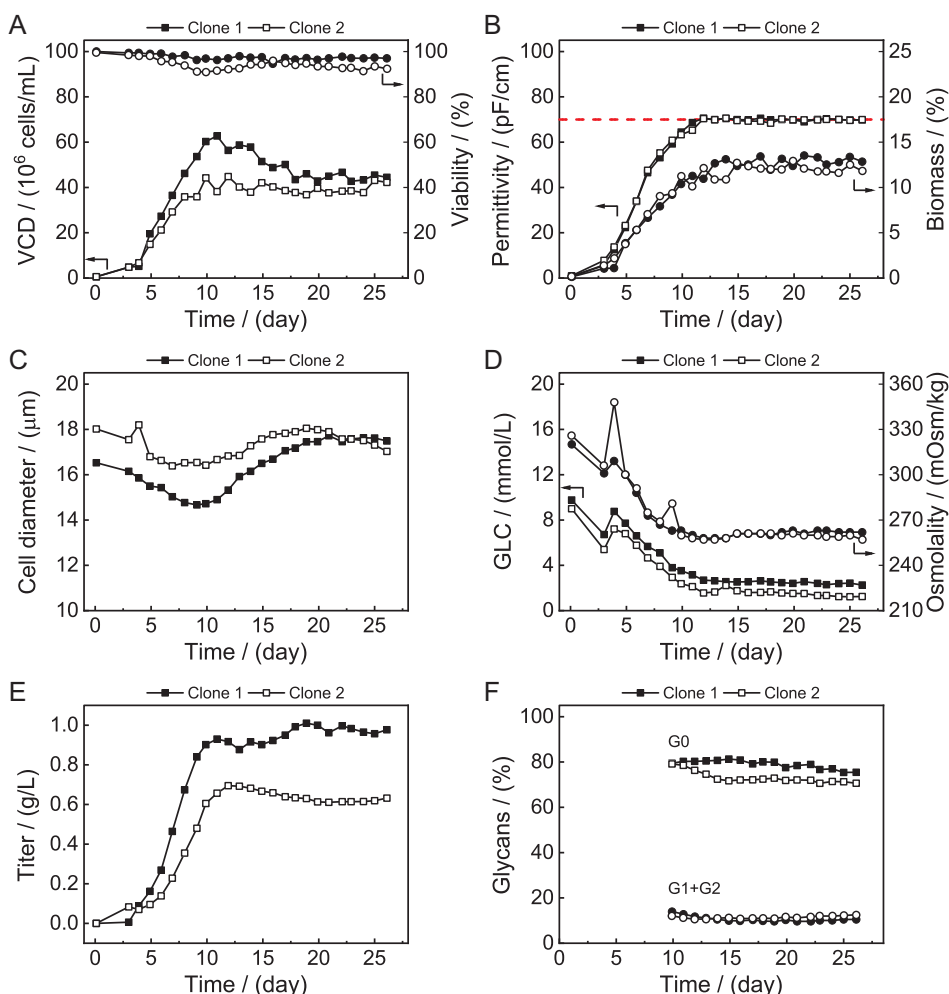


Figure 3.11 Long-term perfusion run at constant permittivity set point for two different clones (■ Clone 1, □ Clone 2) : (A) VCD and viability, (B) Permittivity, permittivity set point (red dashed line) and calculated biomass using Equation (3.4), (C) Cell diameter, (D) Glucose concentration and Osmolality, (E) titre and (F) Glycans G0 and (G1+G2) as a function of time. © Reprinted from *Biotechnology Progress*, Jean-Marc Bielser, Jakub Domaradzki, Jonathan Souquet, Herv Broly and Massimo Morbidelli, ‘Semi-continuous scale-down models for clone and operating parameter screening in perfusion bioreactors’, 2019, with permission from John Wiley and Sons

at day 11 until it reached a plateau value of about 17.5 μ m (Figure 3.11C), while for Clone 2, a similar behaviour was observed, with a minor oscillation of the cell diameter between 16.5 and 17 μ m. In both cases, as shown in Figure 3.11B, the computed value of the reactor biomass remained substantially constant, as expected from the constant values of the controlled permittivity.

Interestingly, the values of glucose concentration, osmolality, mAb concentration and product quality (i.e., glycan patterns) in Figure 3.11D–F remain constant during the entire culture after day 11. This shows that for both clones, the total biomass-based control succeeded in keeping stable the bioreactor steady state, independently of the behaviour of the VCD, which was, in fact, clone specific.

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The preceding experiments show, in particular, that we have to distinguish between at least two types of cell behaviours. In the first one, as for Clone 2, the cell diameter is independent of the state of the cell or shows only slight changes after reaching the targeted permittivity/biomass set point and entering the production phase. In this case, no differentiation between viable cell density and biomass is necessary, and the operation at steady state by total biomass control is possible. In the second type of behaviour, as for Clone 1, the diameter is still changing, typically increasing for some days after the total biomass has reached the stationary phase. As a result, constant biomass does not imply constant VCD, and then steady-state operation becomes questionable. However, at least for the case investigated here, operation at constant biomass still enables constant metabolite and mAb levels, as well as product quality (Figure 3.11D–F), independently of the changes in the VCD value. Nevertheless, the fact that the cell diameter is changing indicates that steady-state conditions have not actually been achieved since some evolution in the cell metabolism is still apparently taking place.

Raman Spectroscopy. Raman spectroscopy is based on the inelastic scattering of monochromatic light directed at the analyte, resulting in a wavelength shift. Although only a small portion of the photons are inelastically scattered – i.e., about one out of 10^9 or 10^{10} – these carry a lot of information and, therefore, provide a true spectral fingerprint of the analyte that can be used for identification and quantification (Buckley & Ryder 2017). Typically, a Raman spectrograph includes three components: a laser, which provides a high-intensity monochromatic source of light, several sampling optics and a detector enabling the acquisition of Raman spectra (Atkins et al. 2010, Esmonde-White et al. 2017).

Raman spectroscopy is a non-destructive and *in situ* analytical method that does not require sampling. It is a robust technique that, in principle, may enable monitoring multiple quantities at the same time. Moreover, it provides important online process information. This last characteristic is probably the most attractive one, also, in the case of mammalian cell cultures.

The main difficulty in using this sensor is related to the extraction of the desired information from the measured spectra. In a clean sample, with few and simple molecular structures, distinct peaks can be observed in the spectra, and each can be directly related to the sample composition. However, in complex samples, such as in the case of a cell culture, which contains a large number of Raman active chemical and biological species, the measured spectra largely consist of very broad envelopes resulting from the overlapping of the peaks corresponding to the many involved species. It is therefore necessary to reduce the high content of spectral noise and emphasise the information content. For this, various pretreatment and modelling tools based on different statistical approaches have been developed (Gautam et al. 2015). Probably the most common one is the partial least square (PLS) regression, which is described in more detail in Chapter 6. A crucial point is the calibration step, where a suitable model correlating the measured spectra with the sample composition is developed and calibrated using a proper set of offline measurements. This model is subsequently used to predict from measured spectra the composition of new samples. The selection of suitable data sets with sufficient variability is fundamental to obtain a reliable predictive model due to the intrinsically limited extrapolation capability of statistical models (Mehdizadeh et al. 2015, Opel et al. 2010, Webster et al. 2018). For example, if a model is calibrated with a specific pair of cell line and medium,

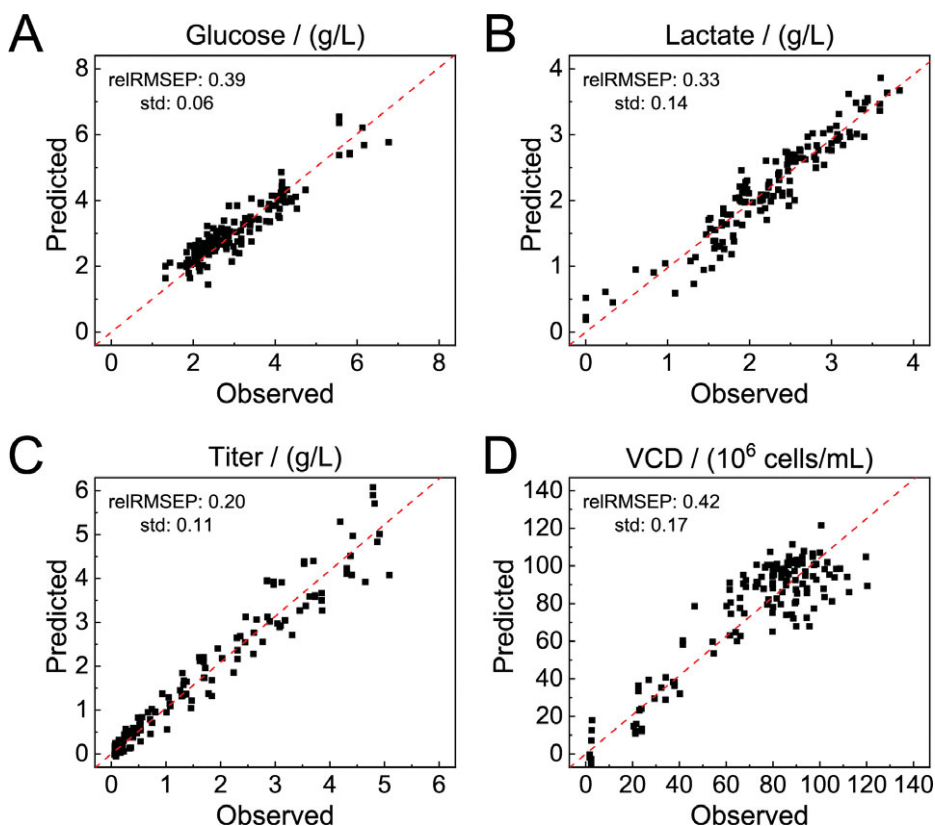


Figure 3.12 Correlation between Raman-based predictions using a PLS model and offline measurements for glucose (A), lactate (B), titre (C) and VCD (D).

it would not necessarily work for a different combination of cell line and medium. This is clearly a limitation of this technology, especially in a process-development environment that is exposed to a large variety of cell lines and media.

Nevertheless, there are several successful implementations of Raman technology reported in the literature. It has been used not only for raw material and media characterisation but also for monitoring and controlling various parameters during cell culture (Buckley & Ryder 2017, Goh et al. 2017). These include the concentration of various metabolites such as glucose, glutamine, glutamate, lactate and ammonia as well as the viable/total cell density (Abu-Absi et al. 2011, Webster et al. 2018, Whelan et al. 2012). With respect to process-control applications, Berry et al. (2016) adjusted the semi-continuous glucose feeding rate based on the online glucose measurement through a Raman sensor, with the aim of reducing undesired glycation of the produced protein. Matthews et al. (2016) used Raman spectroscopy to monitor lactate and glucose levels, and adjusted the glucose feed in order to keep lactate below a certain level while avoiding total glucose depletion. Using this method, the culture could be significantly prolonged, reaching a final titre increase by 85 per cent.

In order to illustrate the potential of this technology, the Raman-based predictions of glucose and lactate concentrations, titre and viable cell density in a representative industrial mammalian cell culture are compared in Figure 3.12 as a function of the corresponding quantities measured by classical offline analytical

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techniques. In this comparison, PLS-based models were used after training on a small but representative data set of six different fed-batch runs. Five runs were used for cross-validation while the sixth one was predicted, resulting in a certain relative root mean square error in prediction (relRMSEP). This was repeated through a rotational approach so that each batch run was predicted once. The obtained results are also shown in the same figure in terms of the average of the errors given by the different rotations (av-relRMSEP) and the corresponding standard deviation (STD). It can be seen that the predictions nicely group around the diagonal going through the origin, with a slight scattering, which is maximal for VCD. The obtained comparison is indeed satisfactory and proves that Raman spectroscopy has, indeed, the potential to provide online several relevant information about the behaviour of the process. It is worth mentioning that the accuracy and robustness of the model predictions can in general be improved by adding more and suitably selected data to the calibration set (Feidl 2019, Feidl et al. 2019).

A very attractive application of Raman spectroscopy in perfusion cultures could be to define online whether steady-state conditions have been achieved or not, due to its sensitivity to small variations in the bioreactor composition. A principle component analysis (PCA), discussed in Chapter 6, could be very useful in this context to quantify small spectral variations and indicate measurement patterns and process evolution. Hence, small drifts from steady state could be detected, and a proper control action could be triggered. Other potential applications include the control of the inlet and outlet reactor streams, such as the bleed by monitoring the VCD, or the control of an independent glucose feed pump dedicated to maintain very low levels of glucose concentration. The possibility of using a single spectrum, and therefore a single probe to reliably monitor several parameters at the same time, makes Raman a very promising technology that we are going to see used more and more often in process monitoring.

Near- and Mid-Infrared Spectroscopy (NIR and MIR). Infrared spectroscopy is another technique used for the quantification of multiple attributes in cell culture processes (Vojinović et al. 2006). In particular, near (NIR) and mid-infrared (MIR) refer to infrared spectroscopy operated in the wavelength ranges of 0.75–1.4 μm and 8–15 μm , respectively. NIR is easier and less costly to employ than MIR, but the latter offers a much better resolution. Because of the high absorbance of water, MIR is, nevertheless, more difficult to use.

As for Raman spectroscopy, suitable chemometric models are needed to best correlate the measured spectra with the quantities that have to be measured (Cervera et al. 2009, Scarff et al. 2006, Vojinović et al. 2006). NIR and MIR have been widely used in the literature to estimate a variety of parameters in very different types of cultures such as yeast, fungi and mammalian cells (Cervera et al. 2009). Finn et al. (2006) reported the use of NIR for the estimation of biomass, glucose, ethanol and protein content in a yeast-based process, whereas Yeung et al. (2002) used the same technology to monitor a flocculation process. Raw material screening for cell culture media was also reported (Kirdar et al. 2010).

Monitoring mammalian cell cultures using NIR or MIR is not very common, and it actually has never been demonstrated in perfusion applications. Arnold et al. (2003) and Ducommun et al. (2002) used an *in situ* probe and created suitable models to estimate glucose, lactate, glutamine and ammonia concentrations in a CHO-K1-based

process. The developed calibration technique provided a reliable model for the interpretation of the raw signals but required a substantial set of offline measurements. Also, Roychoudhury et al. (2007) proposed a multiplexed calibration approach for the monitoring of glucose and lactate, again *in situ* and with mammalian CHO cells.

It is to be noted that infrared, until now, has been mostly used for bacterial fermentations, but, similarly to Raman, it also has as strong potential for online monitoring in mammalian cell cultivations.

3.7 CONCLUSION

The renewed interest in perfusion bioreactors in the context of biomanufacturing is still relatively recent, and most of the related ancillary technologies available on the market are relatively old. This explains why the scale-down models and the systems for monitoring and control of perfusion bioreactors are relatively scarce. Indeed, things are rapidly changing, and we observe a rapid growth in the availability of devices of this type, generally referred to as PAT, from various vendors. This applies also to the process of digitalisation, which is expected to enter much more into the bioprocessing sector in the coming future. This will change the way we approach the design and development of industrial perfusion bioreactors today.

In this chapter, we discussed the availability – and, actually, in most cases, the lack – of scale-down technologies for perfusion cultures. The continuous nature of perfusion and the need for cell retention make the design and realisation of equipment in the nanolitre and millilitre scales technically difficult. For this, the semi-continuous operation has been developed to mimic perfusion, at least in a discretised form. Table 1.3 in Chapter 1 summarises the devices that, mostly based on this approach can currently be used as perfusion scale-down models and their respective monitoring and control characteristics.

Online or at least at-line devices for monitoring and controlling perfusion cultures are also becoming more commonly available, in an attempt to move biomanufacturing to the next stage, dominated by automation and digitalisation. Capacitance is currently the most common variable measured online for monitoring cell biomasses. Raman spectroscopy appears to be very promising for monitoring several variable simultaneously – like titre, nutrients, metabolites and possibly some product-quality-related variables like protein aggregates or glycoforms.

Advanced online monitoring techniques are currently mostly applied at the bench scale, but great benefits could also be obtained by using them in automated micro-scale bioreactors, such as those described in this section. Coupled with the existing high level of automation, these could allow driving the process with an even smaller external interaction. Sampling for cell counts, for example, could be avoided, and the perfusion rate or the bleed rate could be controlled, based on the real-time sensor information.